Advanced Techniques in Pharmaceutical Science

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Advanced Techniques in Pharmaceutical Science (A07955)

Short Title:	Adv Techniques in Pharma Sci		
Faculty	Science and Computing		
Department	Science		
Cluster	Pharmaceutical Science		
Author	EMLANDERS		
Credits	10	**Level:**	Advanced

Description of Module / Aims

This module is an in-depth course in advanced analytical and laboratory techniques. This will include additional aspects of organic and physical chemistry synthesis relating to pharmaceutical science, in addition to their appropriate analysis using: Gas Chromatography (GC), Mass Spectrometry (MS), Liquid Chromatography (LC), Capillary Electrophoresis (CE), Atomic Spectroscopy (AS), Scanning Electron Microscopy (SEM) and Nuclear Magnetic Resonance Spectroscopy (NMR).

Programmes

	BSc (Hons) in Applied Chemistry with Quality Management (`WD_SQMCH_B`)	4 / 1 / M
`PHAR-0016`	BSc (Hons) in Pharmaceutical Science (`WD_SPHSC_B`)	4 / 7 / M

Indicative Content

- Advanced organic synthesis; Use of nuclear magnetic resonance spectrometry (NMR), gas chromatography-mass spectrometry (GC-MS),liquid chromatography-mass spectroscopy (LC-MS), infra-red (IR) and UV-kinetics for characterisation and analysis.
- Diode array and advanced high performance liquid chromatography (HPLC) method development
- Chiral separations by GC, LC and CE
- Atomic spectroscopy, GFAAS and associated interferences
- Surface analysis and characterisation of pharmaceutical materials including polymorphism, scanning electron microscopy (SEM), interfacial tensiometers and solid state nuclear magnetic resonance spectroscopy
- Advanced laboratory work associated with the main techniques included in the module

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise advanced analytical instrumentation.
2. Critique advanced GC, LC, MS, CE, SEM and atomic spectroscopy techniques.
3. Design advanced experiments using an appropriate range of techniques in an integrated manner.
4. Interpret complex analytical data and experimental results.
5. Manage computer-based expert systems in chromatographic method development.
6. Comment on the significance of experimental results/data.

Learning and Teaching Methods

- Lectures.
- Laboratory-based practicals.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	30%	1,2,4
Continuous Assessment	70%	
Practical	70%	3,4,5,6

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of advanced analytical science. Lack of competence with regard to associated laboratory skills, practices and instrumentation.

40%-49%: Will be able to show a limited understanding of the fundamentals of advanced analytical science. Will display some ability with regard to associated laboratory skills, practices and instrumentation.

50%-59%: As well as the above, the student will be to show a reasonable understanding of the fundamentals of advanced analytical science and predict some trends in analytical results. Will display reasonable competence with regard to associated laboratory skills, practices and instrumentation.

60%-69%: In addition, the student will demonstrate a more detailed knowledge of advanced analytical science and show ability to analyse problems in this area. Will display very good competence with regard to associated laboratory skills, practices and instrumentation.

70%-100%: All of the above to an excellent level. In addition, the learner will be able to demonstrate a comprehensive knowledge and understanding of advanced analytical science, excellent reasoning skills and well thought out arguments. Will display advanced competence and independence with regard to associated laboratory skills, practices and instrumentation.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	72	
Independent Learning	166	

Essential Material

- "WIT." wos.heanet.ie. www.wit.ie/library/dbs/scidbs.asp
- "WIT." www.chemweb.com/databases/beilstein. www.wit.ie/library/dbs/scidbs.asp
- "WIT." www.sciencedirect.com. www.wit.ie/library/dbs/scidbs.asp
- "chromacademy.com." www.chromacademy.com

Supplementary Material

- Butcher, D.J. and J. Sned. _Practical G__uide to GFAAS_. New York: Wiley, 1998.

- De Hoffmann, E., J. Charette and V. Stroobant. _Mass Spectrometry: Principles and Applications_. 3rd. Chichester: Wiley, 2007.
- Dean, J.R. _Practical Inductively Coupled Plasma Spectroscopy_. New York: Wiley, 2005.
- Grant, D.W. _Capillary Gas Chromatography_. Chichester: Wiley, 1996.
- Lapman, P. and V. Kriz. _Spectroscopy_. 4th. Belmont USA: Brooks & Cole, 2010.
- Snyder, L.R., J.J. Kirkland and J.L. Glajch. _Practical HPLC Method Development_. Chichester: Wiley, 1997.
- Throck Watson, J. and O.D. Sparkman. _Introduction to Mass_ _Spcetrometry__. Instrumentation, Applications and Strategies for Data Interpretation_. 4th. New York: Wiley, 2007.
- Yergey, A.L. _Liquid Chromatography/Mass Spectrometry_. New York: Plenum Press, 1990.

Requested Resources

- Room Type: Chemistry Lab